Importance of Cryptography: HTTP vs HTTPS Using Wireshark

The main objective of this experiment was to understand why cryptography is important when sending sensitive information over a network. I used Wireshark to capture login traffic from an HTTP test website and an HTTPS website and compared how the login information appeared in the captured packets.

1. Tools Used

- Wireshark
- Web browser
- HTTP test login website
- GitHub login page

2. Procedure

A. HTTP Login

First, I opened Wireshark and selected the active network interface. I started the packet capture and opened an HTTP test login website.

I entered dummy login details and submitted the form. After submitting it, I stopped the capture and checked the packets related to the login request.

The HTTP POST request contained the submitted form data. By looking at the packet details, I could see the username and password in readable form under the HTML Form URL Encoded section.

B. HTTPS Login

For the second test, I started a new capture in Wireshark and opened the GitHub login page, which uses HTTPS.

I entered the login details and submitted the form. After stopping the capture, I checked the packets generated during the login.

This time, I could not see the username and password as readable form data. Instead, the traffic appeared as TLS-encrypted communication.

One important point I noticed is that the password is not something I manually encrypt before entering it. I enter it normally in the browser. The encryption is applied when the information is transmitted through the HTTPS connection.

3. Observations

HTTP

In the HTTP capture, the login information was visible directly in the packet.

The captured POST request showed the form data, including the username and password. This is possible because normal HTTP does not encrypt the data sent between the browser and the server.

Therefore, if someone can capture the network traffic, sensitive information such as login credentials can potentially be read.

Figure 1: HTTP login credentials visible in Wireshark

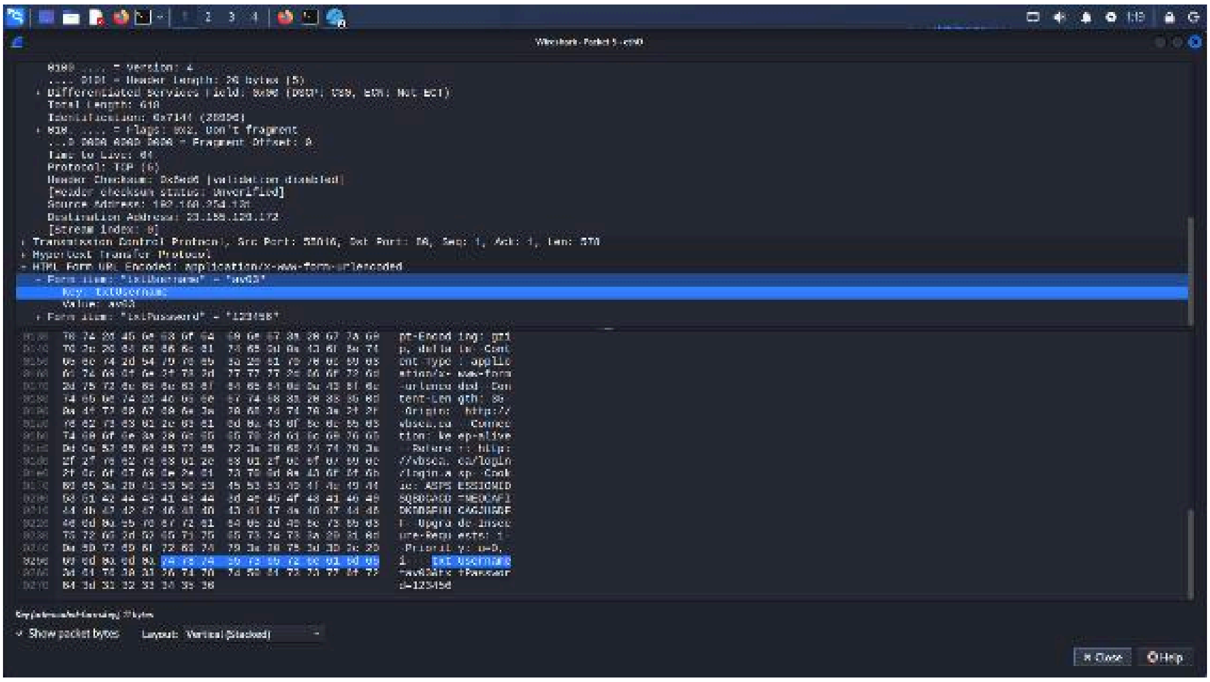

Fig 1.

HTTPS

In the HTTPS capture using GitHub, the communication was protected using TLS (Transport Layer Security).

When I inspected the packets in Wireshark, the actual login information was not visible as readable form data. Instead, the captured packets contained encrypted TLS traffic.

This makes it much harder for someone monitoring the network to obtain the actual login credentials from the captured packets.

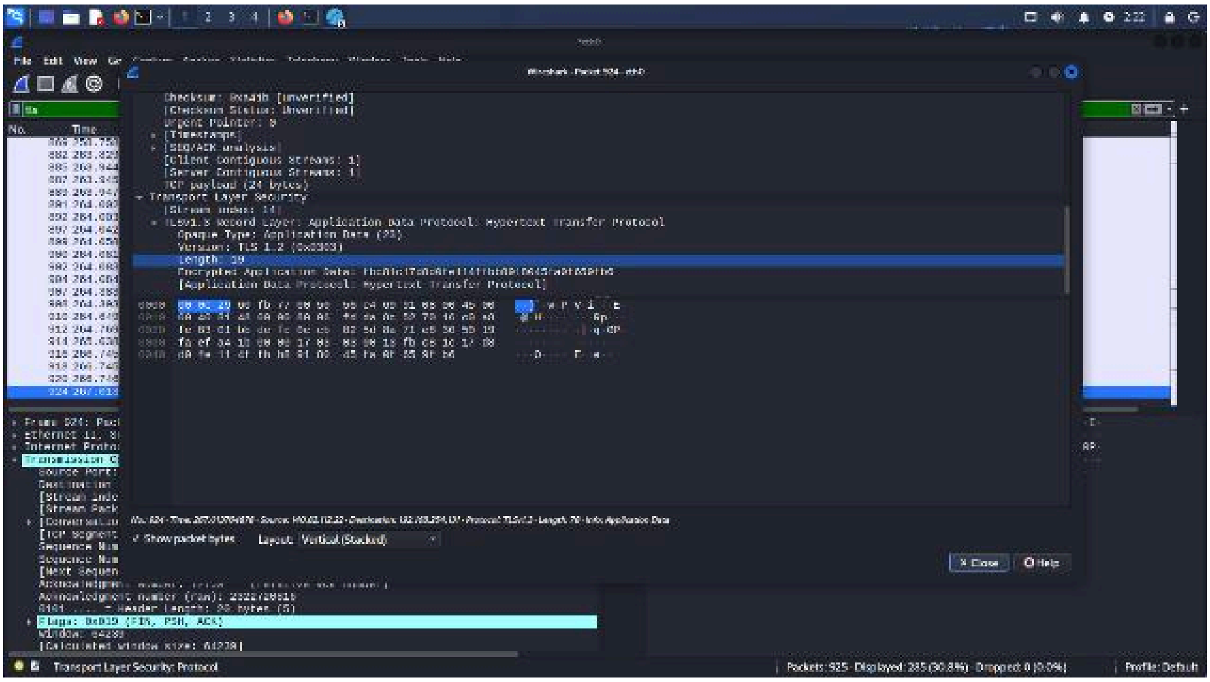

Fig 2

4. HTTP vs HTTPS

HTTP	HTTPS
Data is sent without encryption at the HTTP layer	Data is protected using TLS
Login information can be visible in a packet capture	Login information is not directly readable
Does not provide confidentiality for the transmitted data	Provides confidentiality for the transmitted data
Not suitable for sending passwords securely	Suitable for secure web communication

5. Role of Cryptography

The Wireshark captures made the difference between HTTP and HTTPS easy to understand.

With HTTP, the login details were visible because the data was transmitted without encryption. With HTTPS, the data was protected by TLS, so the actual login information could not be read directly from the captured packets.

Cryptography is therefore important because it protects sensitive information while it is travelling between the user and the server. HTTPS also provides protection against unauthorized modification of the communication and helps verify that the connection is with the intended server.

Conclusion

From this experiment, I was able to see the practical difference between HTTP and HTTPS instead of only understanding it theoretically.

In the HTTP capture, the dummy username and password could be seen directly in Wireshark. In the HTTPS capture, the login traffic was protected by TLS and the credentials were not visible as plain text.

This shows why HTTPS is important for websites that handle passwords and other sensitive information. Using encryption prevents someone who is monitoring the network from simply reading the information being transmitted.

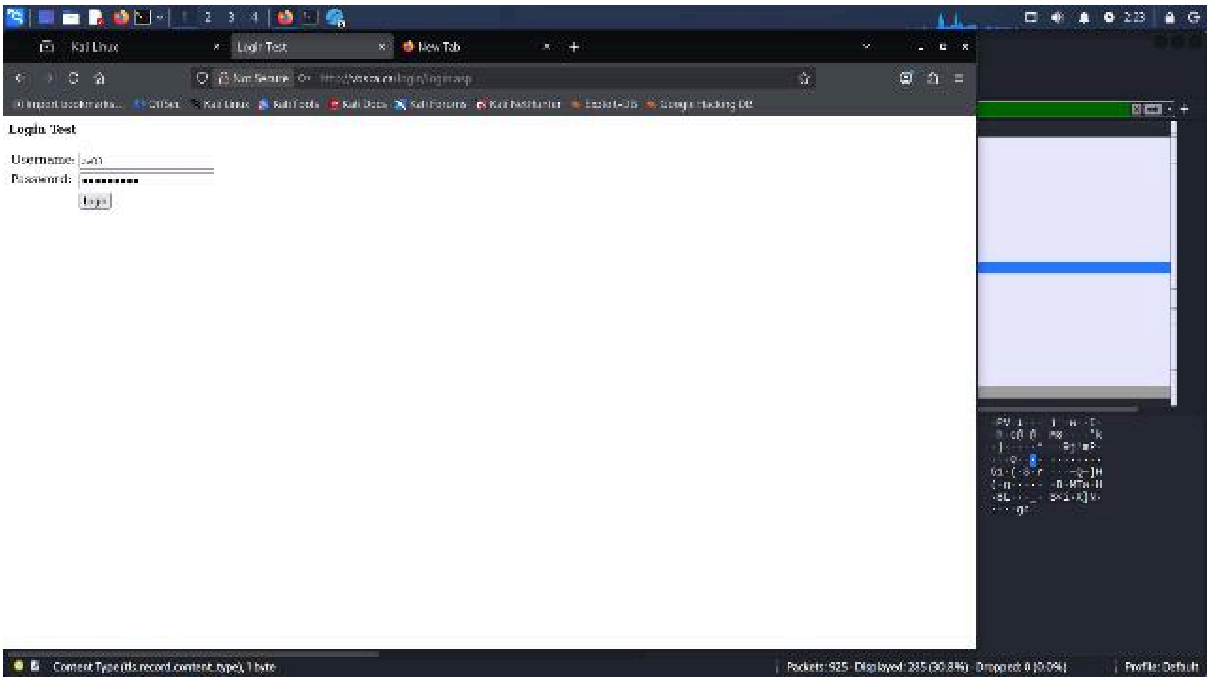

Fig 1.1 Http test login

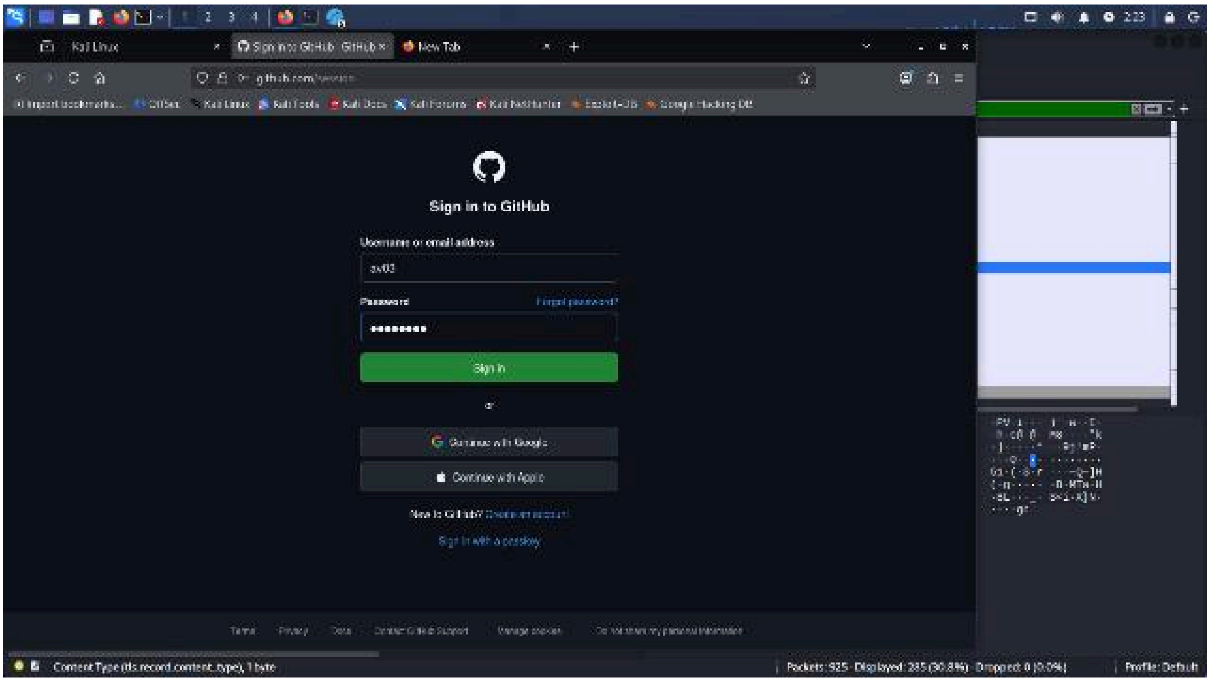

Fig 2.1 Https - Github login